

Static Behavior Evaluation: CMOS Inverter Robustness and Noise Margin

- **Introduction to Noise Margin:**

- Noise margin is the maximum noise voltage a CMOS circuit can tolerate without logic errors.
- If noise amplitude is below the noise margin, it is attenuated through the logic gate, ensuring error-free signal transmission.

- **Function of Noise Margin:**

- Ensures signals with small noise added to **logic 1** remain as logic 1 and those with noise on **logic 0** remain as logic 0.

- **Application:**

- Consider two back-to-back CMOS inverters.
- The voltage transfer curve (VTC) determines noise margins:
 - **Ideal VTC, Piece-wise Linear VTC, and Realistic VTC** depict inverter robustness.

Noise Margin: VIL and VIH Points

- **Definition:**

- **VIL** and **VIH** are operational points where the slope of the voltage transfer curve (VTC) is **-1**.
- These points correspond to the gain of the inverter being equal to **-1**.

- **Logic Recognition:**

- Input voltage **0 to VIL** → Recognized as **Logic 0**.
- Input voltage **VIH to VDD** → Recognized as **Logic 1**.

Noise Margins in CMOS Logic Gates

1. **Conditions for Proper Operation:**

- The following conditions must hold for a logic gate to function correctly in the presence of noise:
- $VOL_MAX < VIL_MAX$
- $VOH_MIN > VIH_MIN$

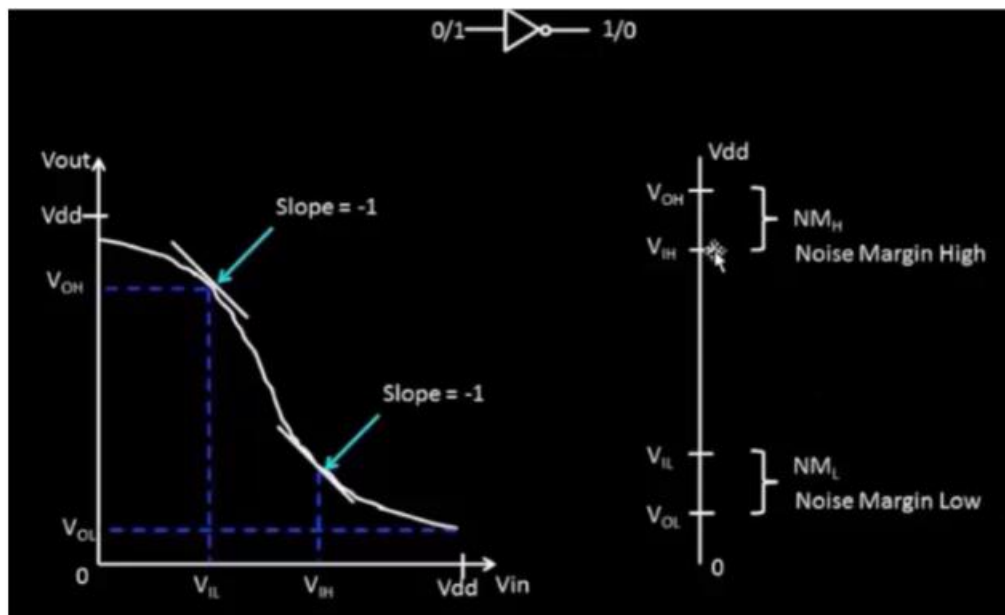
2. **Behavior in Different Input Ranges:**

- **For $V_{in} \leq V_{IL}$:**
The inverter gain magnitude is less than unity, resulting in minimal output change for a given input change.
- **For $V_{in} \geq V_{IH}$:**
The gain magnitude is also less than unity, leading to minimal output change.
- **For $V_{IL} < V_{in} < V_{IH}$:**
The gain magnitude exceeds unity, causing significant output changes. This range is called the **undefined region** because noise signals pushing V_{in} into this range may introduce errors.

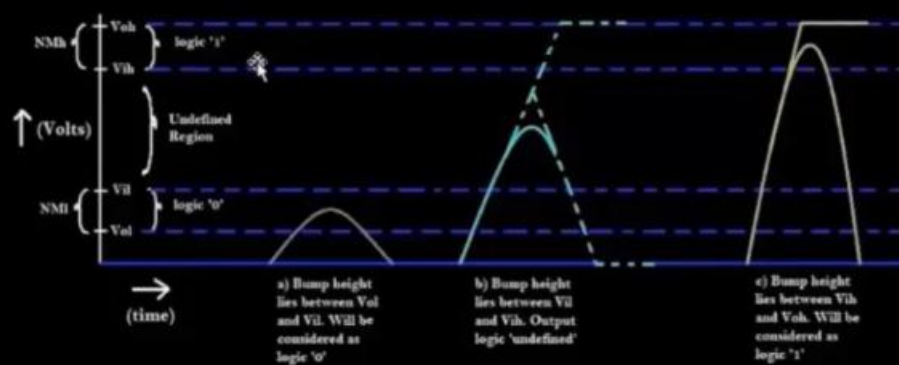
3. Noise Margin Equations:

- **Low-Level Noise Margin (NML):**
- $NML = V_{IL_MAX} - V_{OL_MAX}$
- **High-Level Noise Margin (NMH):**
- $NMH = V_{OH_MIN} - V_{IH_MIN}$
- **Overall Noise Margin (NM):**
- $NM = \text{Min}(NML, NMH)$

These noise margins define the tolerance of a circuit to noise without compromising its logical operation.

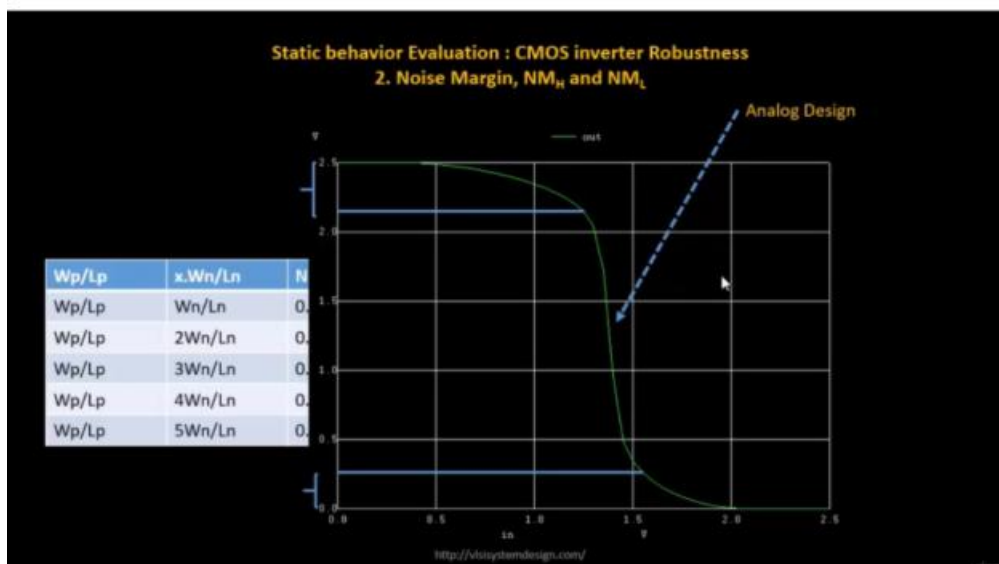
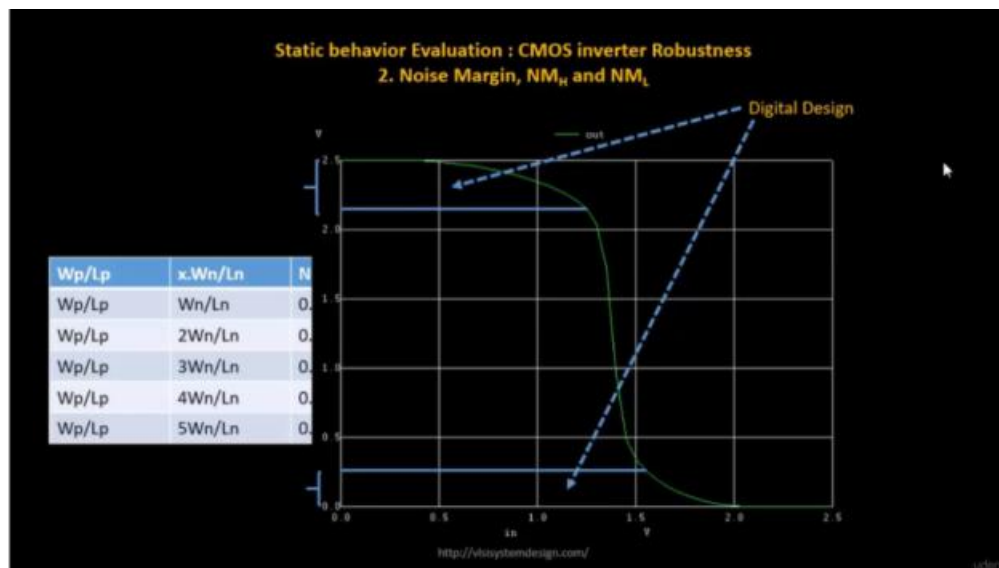


Noise Margin Summary



Noise induced bump characteristics at different noise margin levels

For any signal to be considered as logic '0' and logic '1', it should be in the NM_L and NM_H ranges, respectively



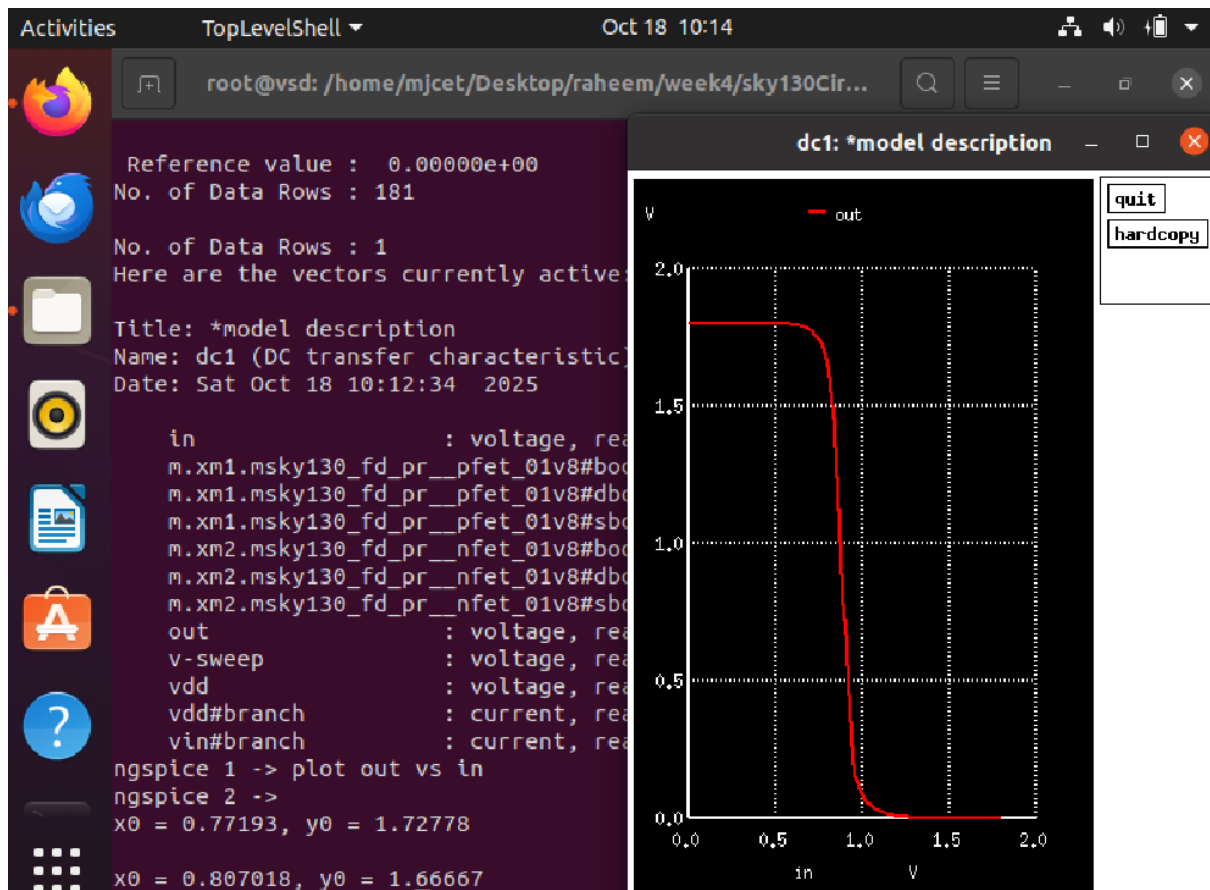
Noise Margin Robustness against variations in Device Ratio

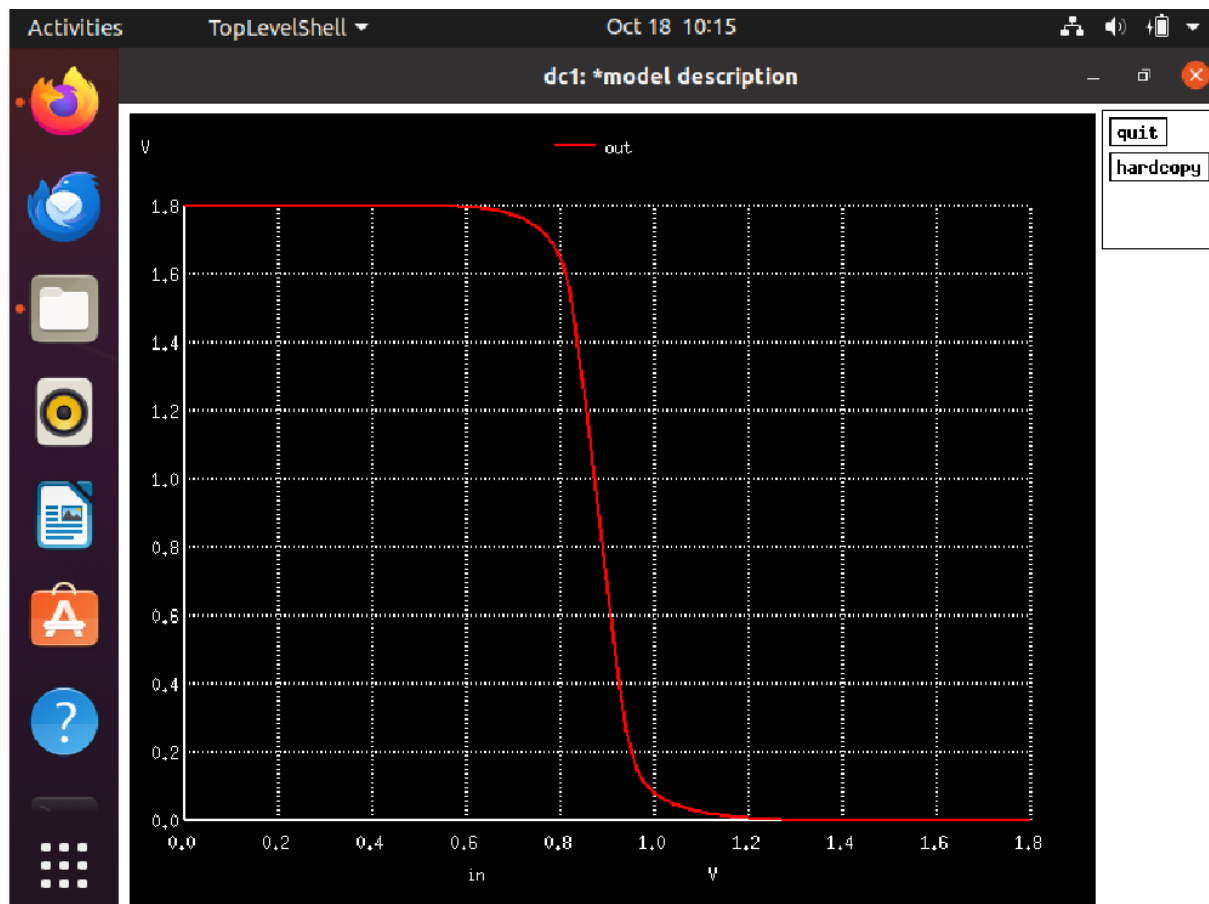


LABS

Static behavior evaluation: CMOS inverter robustness, Noise margin

Noise Margin - sky130 Inverter ($W_p/L_p=1\mu/0.15\mu$, $W_n/L_n=0.36\mu/0.15\mu$)





Thank you VSD Team for mentoring the RSIC V Reference SoC Tapeout Program
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